

Vinaya Vijayan

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SUMMARY

Machine Learning engineer with 6.5+ years of experience in building successful algorithms and predictive models for different industrial sectors. Experience in multiple applications such as Generative AI, LLM, Natural Language Processing, Image Processing, Predictive analytics etc., with a comprehensive understanding of the life cycle, from data collection, visualization to building models to deployment and monitoring

TECHNICAL SKILLS

Programming Languages	Python, SQL
Databases	MySQL, PostgreSQL
Frameworks/Packages	OpenCV, SK-Learn, NLTK, Spacy, Pandas, Keras, Tensorflow, FastAPI, Streamlit
Operating Systems	Windows, Ubuntu
Cloud	AWS Sagemaker, AWS Comprehend, AWS Textract, AWS Bedrock, AWS S3 Kubernetes, Azure
Others	Jira, Git, Docker, Huggingface

EXPERIENCE

LITMUS7 SYSTEMS CONSULTING LTD

October 2022 - Current

Review Summarization using GenAI

- Generated a review summarization engine for an ecommerce platform using Python and Generative AI.
- With LLM and Prompt Engineering for generated relevant topics in positive and negative comments and for summarizing the reviews for each product

Incident Ticket Management System

- The solution has 4 modules. The forecasting module predicts ticket frequency aiding resource allocation and proactive management of incidents. Classification module categorizes tickets into assignment groups and suggests similar tickets for efficient handling. Its ticket similarity search help associates to leverages past resolutions to expedite current issue resolution
- Also has a dynamic dashboard showcasing ticket distribution across various groups.

Derma Scan - Computer Vision

- Solution to identify cosmetic skin conditions from an uploaded or captured image
- Custom trained image processing models were used to identify different skin conditions and highlight them and recommend products based on the condition
- Detectron and opencv was predominantly used for the solution development

Social Media Analytics - Meta, Python, Fivetran

- Created A solution in python that helps the customer to get an insight on social media performance and customer feedback's on the medium.
- Customer will be able to get the information of both Facebook and Instagram medium in the same solution and unlike a meta management page the insights scope goes beyond just reactions, comment count and likes to actual content that is generated in the comment section.

Image Quality Analysis

- Created a classification solution based on OpenCV, Tensorflow which can identify issues with image that is to be uploaded to a retail Ecom platform.

Churn Analytics

- Developed an AI based solution for an Ecom Client to identify the customer churn rate.
- Our solution utilizes customer purchase patterns to identify potential churners and predict their churn timeline based on historical data enabling proactive retention strategies

EMVIGO TECHNOLOGIES

July 2021 - October 2022

Fin-Tech Investor Preference Identification

- Solution to identify the preferred investment sectors and type of investors from their company descriptions and email responses.
- Developed model to identify relevant features of an investor from the different templates of data.
- Trained customized BERT model using for sentiment analysis post comparison with AWS comprehend
- A custom spacy model was used for entity recognition.

Email Invoice Processing - Retail Sector

- Developed a solution that extracts attributes from invoice emails of online shopping platforms.
- Classification based on language identification module was developed using Spacy NLP.
- Machine learning, Deep learning - object detection, Image processing and logical modules were used for extraction.
- Training was done using AWS Sagemaker with annotations done locally to reduce costs

TATA CONSULTANCY SERVICES

2017 October - 2021 June

Mortgage Process Digitisation

- Developed a solution for the classification of Mortgage documents to 16 classes.
- Build a solution to extract class specific attributes from each page of the document.
- Setup Azure Service Bus for pipeline integration.
- Implemented machine learning ,deep learning technologies, NLP techniques.
- Experience in Code - Model deployment to Dev, UAT, Production environment
- Developed and deployed metric to monitor the performance of models.
- Created and deployed CronJob for data management.
- Developed S3 pipelines for data ingestion to azure storage.

Contract Digitisation

- Data collection and analysis of documents
- Build a solution to extract required clauses from unstructured contract documents of customer
- Used machine learning, deep learning technologies, and NLP techniques to extract specific clauses

Legal Invoice Document Attribute Extraction

- Developed a Machine learning solution for the extraction of various attributes from OCR enabled and Disabled PDF legal documents.
- Image processing methods using OpenCV was applied for preprocessing of images and grouping data

Attribute Extraction from International Trade Invoices

- Extraction of various attributes from images using Machine Learning,Deep Learning and Spacy.
- Developed object detection model using Deep Learning technology
- Developed image processing code to enhance textual extraction using tesseract.
- Database administration for the storage, retrieval and manipulation of extracted data.

Data Mining and Attribute Extraction

- An Application to scrape contents from different sites using Selenium and extract different attributes using machine learning, NLTK and Spacy.

Attribute Extraction from Pharmaceutical Images

- Developed a solution to extract medical features from image documents using machine learning, NLTK and Spacy along with Image Processing for image enhancement.

EDUCATION

Adishankara Institute of Engineering and Technology, Kalady
Btech in Computer Science Engineering

Percentage 74%
2013 - 2017

Nirmala School Aluva
Senior Secondary Education

Percentage 90%
2011 - 2013

Nirmala School Aluva
Secondary Education

Percentage 97%
2011

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals

Microsoft

Career Essentials in Generative AI by Microsoft and LinkedIn

Microsoft and LinkedIn

Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning

Coursera

Convolutional Neural Networks in TensorFlow

Coursera

Coursera Natural Language Processing in TensorFlow

Coursera

REFERENCES

Vivek Suresh

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